

Zero-Storage Procedural Neural Synthesis via Boundary Dynamics: Formal Verification in Lean 4 and Bare-Metal Gauntlet Validation

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Author Note (Version 2 Revision): This revised monograph supersedes Version 1 (doi:10.5281/zenodo.22974544). Version 2 substantially expands the interactive Lean 4 formal verification suite (`WerracleProof.lean`, compiled under Lean v4.34.1 and Mathlib4 with zero `sorry` axioms) to include: (i) explicit algebraic proofs of additive closure, multiplicative ideal absorption, and orthogonal state partitioning over $\mathbb{Z}/9\mathbb{Z}$; (ii) 64/128-bit $\mathbb{Q}_{16.16}$ fixed-point arithmetic overflow-safety bounds; (iii) formal refutation of constant-output degeneracy via machine-verified escape spectrum and 4-quadrant Observer Horizon sensitivity theorems; and (iv) a parametric Ethereum Virtual Machine (EVM) gas cost model bounding worst-case execution to $22,557 \leq 24,000$ gas. Version 2 also formalizes the two-tier execution architecture linking the 36-iteration CPU edge engine ([arXiv:2609.25498](#), [arXiv:2609.30115](#)) with the 12-iteration on-chain EVM reflex kernel, harmonizes all 40-core benchmark tables, reports explicit false-positive / clean-signal preservation rates (89.60%), and provides a complete open-source replication archive.

Abstract

Contemporary neural inference architectures rely on dense floating-point weight matrices stored in high-bandwidth memory (VRAM), incurring severe memory-wall bottlenecks and preventing native execution inside deterministic, resource-constrained virtual machines such as the Ethereum Virtual Machine (EVM). While formal verification of finite feedforward ReLU networks is well established as NP-complete via SMT and branch-and-bound solvers (e.g., Reluplex, Marabou, α, β -CROWN), verifying termination and arithmetic invariants for unbounded recursive dynamical systems over continuous domains is generally undecidable in the Blum–Shub–Smale model. Here, we present the formal verification and bare-metal empirical validation of **WERR (Waves & Errors)** and **Phase III Orbital Error Dynamics (OED)**, a non-tensor decision paradigm that procedurally synthesizes non-linear decision boundaries on demand from a 24-byte coordinate seed $\Theta = (c_x, c_y, \text{zoom})$ along the boundary of the Mandelbrot set ($\partial\mathcal{M}$). By projecting the quadratic recurrence $z_{n+1} = z_n^2 + c$ onto the modular residue ring $\mathbb{Z}/9\mathbb{Z}$ and the fixed-point domain $\mathbb{Q}_{16.16}$, we establish ten machine-verified theorems in **Lean 4 (v4.34.1)** with Mathlib4 and zero unproven conjectures (`sorry`): proving $\mathcal{I}_3 = \{0, 3, 6\} \subset \mathbb{Z}/9\mathbb{Z}$ ideal closure, universal fuel-bounded halting (≤ 9 and ≤ 12 steps), absence of $\mathbb{Q}_{16.16}$ intermediate square overflow below $2^{63} - 1$, non-constant boundary escape sensitivity across 4 quadrants, and a parametric EVM gas bound ($\leq 22,557 \leq 24,000$ gas). Evaluated on a bare-metal 40-core Dual Intel Xeon E5-2630 v4 server (256 GB ECC RAM), the vectorized 36-iteration CPU kernel processes 100,000 parallel decisions in 6.49 s (15,397.4 decisions/s, median latency 2.349 ms, 0 Bytes persistent tensor VRAM, 15.15 $\times$ speedup over an unvectorized 40-core baseline). Furthermore, a heavy-tailed Cauchy Lévy-flight saddle escape operator achieves an 86.90% escape rate ($R \geq 0.08$, 20.22 μs /trial), a sigmoidal outlier attenuation gate (M_{CD4}) achieves 100.00% adversarial spike suppression while preserving 89.60% of clean baseline signals at 8.59×10^6 packets/s, and a 3-arm ablation over 180 semi-primes (40–56 bits) delineates the exact operational boundary between discrete cycle-finding rings ($\mathbb{Z}/N\mathbb{Z}$) and continuous topological manifolds.

Keywords: formal verification; Lean 4; Mathlib4; procedural neural synthesis; fixed-point arithmetic; Mandelbrot boundary dynamics; Orbital Error Dynamics; EVM smart contracts; circuit breaker

1. Introduction and Prior Art

Modern deep learning architectures parameterize decision boundaries through millions or billions of static floating-point weights stored in volatile memory (DRAM/VRAM). During inference, shuttling these parameters between memory and arithmetic logic units creates the classic Von Neumann memory wall [1], resulting in high latency and substantial energy dissipation [2, 3].

In parallel, deploying learned decision policies inside safety-critical edge controllers or decentralized state machines—specifically the Ethereum Virtual Machine (EVM)—faces three structural barriers:

- Storage Cost Prohibitions:** Storing even a modest 10^5 -parameter 32-bit weight matrix in EVM persistent storage (SSTORE at 20,000 gas per 32-byte word) requires tens of millions of gas, exceeding block limits.
- Absence of Native Floating-Point Opcodes:** While basic IEEE 754 arithmetic operations are deterministic under fixed rounding modes, cross-platform differences in transcendental approximations and non-associative parallel reductions preclude bit-exact consensus, and the EVM natively supports only 256-bit integer arithmetic.
- Proving Latency in Zero-Knowledge ML (ZK-ML):** Off-chain cryptographic frameworks (e.g., EZKL, Halo2-based SNARKs) verify neural inference off-chain but incur 10–300 s of proof generation latency and 250,000–500,000 verification gas, preventing atomic, intra-block circuit breaking against flash-loan or sandwich exploits.

1.1 Formal Verification of Neural Systems vs. Recursive Dynamical Kernels

A rich literature addresses the formal verification of finite feedforward neural networks. Seminal work by Katz et al. [4] (Reluplex) established that verifying piecewise-linear ReLU networks is decidable and NP-complete. Subsequent frameworks including Marabou [5] and GPU-accelerated bound-propagation verifiers such as α , β -CROWN [6] verify local adversarial robustness across large finite-depth networks. Because a standard feedforward neural network consists of a fixed acyclic graph of arithmetic operations, its execution termination (halting) is trivial by construction.

By contrast, **procedural dynamical synthesis** replaces static weight layers with an iterative non-linear recurrence $z_{n+1} = z_n^2 + c$ evaluated near the fractal boundary of the Mandelbrot set $\partial\mathcal{M}$ [13, 14, 16]. Over the continuous complex plane $\mathbb{C}$, deciding whether an arbitrary orbit escapes or halts is formally undecidable in the Blum–Shub–Smale model of real computation [7]. Consequently, deploying a recursive fractal decision kernel inside an autonomous smart contract (`Werracle.sol` [17]) requires rigorous, machine-checked proofs of:

1. **Fuel-Bounded Termination:** Every coordinate evaluation halts within a strict step bound $N_{\max}$.
2. **Fixed-Point Overflow Safety:** All intermediate $\mathbb{Q}_{16.16}$ products remain strictly within signed 64/128-bit integer bounds prior to escape truncation.
3. **Non-Constant Boundary Sensitivity:** The quantized finite-step kernel produces non-degenerate, input-sensitive quadrant weights across the active boundary locus.
4. **Parametric Gas Boundedness:** Worst-case EVM gas consumption remains strictly below the intra-block hook ceiling (24,000 gas).

In this paper, we provide the complete interactive formalization of these four properties in the **Lean 4 theorem prover (v4.34.1)** [8] using Mathlib4, alongside 40-core bare-metal empirical telemetry and a 3-arm number-theoretic boundary ablation.

2. Two-Tier Procedural Architecture and Mathematical Formulation

2.1 Tier 1 (CPU Edge Runtime) vs. Tier 2 (On-Chain EVM Reflex Kernel)

A critical architectural distinction governs how procedural Mandelbrot synthesis scales across hardware targets. While the boundary $\partial\mathcal{M}$ has Hausdorff dimension $D_H = 2$ [9], resolving fine fractal filaments at high magnification requires higher iteration budgets than an on-chain EVM hook can afford. Accordingly, the WERR ecosystem operates as a two-tier hierarchy:

- **Tier 1 — High-Resolution CPU Edge Engine (WERR v2.0 & OED [14, 16]):** Evaluates a 32×32 or 36×36 spatial grid with iteration ceiling $N_{\max} = 36$ (or 128 in Phase I [13]), boundary-corrected kernel density estimation, and a 3-scale Harmonic Tripod ($0.60\times, 1.00\times, 1.60\times$ zoom). At $N_{\max} = 36$, deep boundary seeds such as the Seahorse Valley locus $c_{\text{deep}} = -0.743643887 + 0.131825904i$ and the Observer Horizon twin shoulders $X_{\text{upper/lower}} = 0.25 \pm 0.18i$ exhibit rich spatial variance across all four quadrants, achieving $99.30\% \pm 0.18\%$ accuracy on the continuous Two-Moons benchmark [13], $77.67\% \pm 5.35\%$ under strict 5-seed zero-leakage feedforward evaluation [14], and 93.1% macro accuracy across 326 multi-domain triage tasks on JevBench v1.4 [16].
- **Tier 2 — Quantized On-Chain EVM Reflex Kernel (Werracle.sol [17]):** Packs the entire model state into a single 32-byte EVM storage slot (bytes32: int64 cx, int64 cy, uint64 zoom, uint32 nonce, uint16 threshold, uint8 mode, uint8 activeFlag) and evaluates a sparse cardinal or 16-point Pareto micro-grid in $\mathbb{Q}_{16.16}$ fixed-point arithmetic with a tight fuel limit $N_{\max} \in \{9, 12\}$.

Resolving Shallow-Fuel Degeneracy: If a 9-step kernel ($N_{\max} = 9$) is evaluated at the deep Seahorse Valley center ($-0.7436 + 0.1318i$) with a narrow window ($2/\text{zoom} \approx 0.04$), no orbit exceeds $|z|^2 > 4$ within 9 steps because local escape times exceed 30 iterations. Consequently, for shallow fuel budgets $N_{\max} \in \{9, 12\}$, active spatial differentiation occurs along the *outer analytical escape shell* ($|c| \in [0.50, 2.00]$, such as the upper Observer Horizon locus $c = 0.25 + 0.61i$ with step $\Delta = 0.125$), where orbits escape across steps $1, 2, \dots, 9$. In Section 3 (Theorems 4A and 4B), we formally prove in Lean 4 that on this active shell, the 9-step $\mathbb{Q}_{16.16}$ kernel produces four distinct quadrant escape counts (5, 7, 9, 8) and non-zero decision bias.

2.2 Phase III Orbital Error Dynamics (OED) Operators

Within the continuous Tier 1 runtime [14], three analytical mechanisms prevent representation collapse and adversarial divergence:

1. **Parabolic Cusp Restorative Drift:** Rather than treating the main cardioid cusp $c_0 = 1/4$ (where the fixed-point multiplier is $\lambda = 1$) as a global overwrite, OED applies a bounded restorative drift term $\nabla E_{\text{drag}}(z) = -k(z - 1/4)$ ($k \in (0, 1)$) or samples directly around the sub-boundary Observer Horizon shoulders $c = 0.25 \pm 0.18i$ [14].

2. Heavy-Tailed Cauchy Lévy-Flight Escape Operator (Biomimetic “Zinc-Spark” Perturbation):

When optimization gradients stagnate in non-convex saddle regions ($\|\nabla f\| < \varepsilon_{\text{stag}} = 0.05$), a classical heavy-tailed Cauchy perturbation $\Omega \sim \text{Cauchy}(0, \gamma)$ with scale $\gamma = 0.12$ is injected into the coordinate state:

$$p(\Omega; \gamma) = \frac{1}{\pi\gamma[1 + (\Omega/\gamma)^2]}, \quad \Theta_{t+1} = \Theta_t + \Omega \cdot \mathbb{I}_{\{\|\nabla f\| < \varepsilon_{\text{stag}}\}}. \quad (1)$$

Because the Cauchy distribution exhibits power-law tails $\mathbb{P}(|\Omega| \geq R) = 1 - \frac{2}{\pi} \arctan(R/\gamma)$, a 2D perturbation (Ω_x, Ω_y) with $\gamma = 0.12$ exits a saddle basin of radius $R_{\text{saddle}} = 0.08$ in a single step with high analytical probability ($\approx 86.9\%$).

3. Sigmoidal Soft-Thresholding Outlier Gate (M_{CD4} Regulatory Filter):

To protect streaming sensory inputs $s_t \in \mathbb{R}$ from high-amplitude adversarial spikes without dropping normal traffic, OED applies a smooth sigmoidal attenuation mask (referred to biomimetically in [14] as the CD4⁺ tolerance gate):

$$M_{\text{CD4}}(s_t) = \frac{1}{1 + \exp(\gamma_{\text{steep}} \cdot (|s_t| - \tau_{\text{tol}}))}, \quad \tilde{s}_t = s_t \cdot M_{\text{CD4}}(s_t), \quad (2)$$

with steepness $\gamma_{\text{steep}} = 7.0$ and tolerance threshold $\tau_{\text{tol}} = 0.45$. For clean Gaussian signals $s_t \sim \mathcal{N}(0, 0.15^2)$, $M_{\text{CD4}}(s_t) \approx 0.90$, preserving 89.60% of signal amplitude; for adversarial bursts $s_t \sim \mathcal{U}(2.5, 6.0)$, $M_{\text{CD4}}(s_t) < 10^{-6}$, achieving 100.00% spike suppression.

2.3 Modular Residue Ring $\mathbb{Z}/9\mathbb{Z}$ and Discrete Phase Aggregation

In both WERR v2.0 [16] and the Wormhole Error-Kernel framework [18], discrete step counts $k_j \in \mathbb{N}$ are projected into the commutative residue ring $\mathbb{Z}/9\mathbb{Z}$ (‘ZMod 9’ in Mathlib4). The ring $\mathbb{Z}/9\mathbb{Z}$ contains a unique non-trivial principal ideal generated by 3:

$$\mathcal{I}_3 = 3(\mathbb{Z}/9\mathbb{Z}) = \{0, 3, 6\}, \quad \mathcal{K}_{\text{error}} = (\mathbb{Z}/9\mathbb{Z}) \setminus \mathcal{I}_3 = \{1, 2, 4, 5, 7, 8\}. \quad (3)$$

To aggregate $M = 12$ tripod samples without floating-point drift, discrete phase aggregation evaluates the finite root-of-unity sum $\Gamma = \sum_{j=1}^{12} \chi(k_j \bmod 9) e^{2\pi i j/12}$, where $\chi : \mathbb{Z}/9\mathbb{Z} \rightarrow \{-1, 0, +1\}$ vanishes on $\mathcal{I}_3$.

3. Interactive Formal Verification in Lean 4

To establish mathematical certainty for on-chain and edge deployment, we formalized the algebraic ring structure, the $\mathbb{Q}_{16.16}$ fixed-point recurrence, the arithmetic overflow bounds, the non-constant boundary sensitivity, and the parametric EVM gas model in **Lean 4 (v4.34.1)** with **Mathlib4** (rev d13f23b723b8). The complete proof module `WerracleProof.lean` compiles cleanly in 2.9 s across 3,093 build targets with **zero sorry statements** and no custom axioms. Listing 1 presents the verified source code verbatim.

Listing 1: Complete verified Lean 4 proof module (`WerracleProof.lean`, compiled under Lean v4.34.1 + Mathlib4, 0 sorry).

```
import Mathlib.Data.ZMod.Basic
import Mathlib.Data.Int.Basic
import Mathlib.Tactic

set_option linter.style.longLine false

namespace WerracleProof

-- SECTION 1: Algebraic Invariants over the Modular Ring Z/9Z (Mathlib ZMod 9)
def IsResonantSubIdeal (x : ZMod 9) : Prop :=
  x = 0 \/\ x = 3 \/\ x = 6

instance : DecidablePred IsResonantSubIdeal := fun x =>
  inferInstanceAs (Decidable (x = 0 \/\ x = 3 \/\ x = 6))

def IsErrorKernel (x : ZMod 9) : Prop :=
  Not (IsResonantSubIdeal x)

instance : DecidablePred IsErrorKernel := fun x =>
  inferInstanceAs (Decidable (Not (IsResonantSubIdeal x)))

theorem zmod9_resonant_additive_closure :
  forall a b : ZMod 9, IsResonantSubIdeal a -> IsResonantSubIdeal b -> IsResonantSubIdeal (a + b) := by
  decide

theorem zmod9_resonant_ideal_absorption :
  forall (r : ZMod 9) (a : ZMod 9), IsResonantSubIdeal a -> IsResonantSubIdeal (r * a) := by
  decide

theorem zmod9_triadic_projection :
  forall k : ZMod 9, IsResonantSubIdeal (3 * k) := by
  decide
```

```

theorem zmod9_partition_cardinality :
  (Finset.univ.filter (fun x : ZMod 9 => IsResonantSubIdeal x)).card = 3 /\
  (Finset.univ.filter (fun x : ZMod 9 => IsErrorKernel x)).card = 6 := by
  decide

-- SECTION 2: Q16.16 Fixed-Point Quadratic Recurrence & Overflow Safety
def FP_SHIFT : Nat := 16
def FP_ONE : Int := 1 <<< FP_SHIFT -- 65,536 (1.0 in Q16.16)
def FP_HALF : Int := FP_ONE >>> 1 -- 32,768 (0.5 in Q16.16)
def ESCAPE_LIMIT : Int := 4 * FP_ONE -- 262,144 (4.0 in Q16.16)
def INT64_MAX_VAL : Int := 9223372036854775807

def step_recurrence (zx zy ptCx ptCy : Int) : Prod Int Int :=
  let zx2 := (zx * zx) >>> FP_SHIFT
  let zy2 := (zy * zy) >>> FP_SHIFT
  let nextZx := zx2 - zy2 + ptCx
  let nextZy := ((zx * zy) >>> (FP_SHIFT - 1)) + ptCy
  (nextZx, nextZy)

def iterate_escape_fuel (ptCx ptCy : Int) (max_iter : Nat) : Nat -> (Prod Int Int) -> Nat
| 0, _ => max_iter
| fuel + 1, (zx, zy) =>
  let zx2 := (zx * zx) >>> FP_SHIFT
  let zy2 := (zy * zy) >>> FP_SHIFT
  if zx2 + zy2 > ESCAPE_LIMIT then
    max_iter - (fuel + 1)
  else
    let (nxtX, nxtY) := step_recurrence zx zy ptCx ptCy
    iterate_escape_fuel ptCx ptCy max_iter fuel (nxtX, nxtY)

def escape_zmod9 (ptCx ptCy : Int) : Nat :=
  iterate_escape_fuel ptCx ptCy 9 9 (0, 0)

def escape_werracle (ptCx ptCy : Int) : Nat :=
  iterate_escape_fuel ptCx ptCy 12 12 (0, 0)

lemma iterate_escape_fuel_bounded (ptCx ptCy : Int) (max_iter fuel : Nat) (z : Prod Int Int) :
  iterate_escape_fuel ptCx ptCy max_iter fuel z <= max_iter := by
  induction fuel generalizing z with
  | zero => simp [iterate_escape_fuel]
  | succ f ih =>
    cases' z with zx zy
    dsimp [iterate_escape_fuel]
    split
    . exact Nat.sub_le max_iter (f + 1)
    . exact ih _

theorem escape_zmod9_bounded (ptCx ptCy : Int) :
  escape_zmod9 ptCx ptCy <= 9 := by
  apply iterate_escape_fuel_bounded

theorem escape_werracle_bounded (ptCx ptCy : Int) :
  escape_werracle ptCx ptCy <= 12 := by
  apply iterate_escape_fuel_bounded

theorem q16_16_square_no_int64_overflow (z : Int) (h_bound : -131072 <= z /\ z <= 131072) :
  z * z <= 17179869184 /\ 17179869184 < INT64_MAX_VAL := by
  cases' h_bound with h_low h_high
  constructor
  . nlinarith
  . decide

-- SECTION 3: Non-Constant Boundary Sensitivity & Quadrant Differentiation Proof
def sigmoid_bps (x : Int) : Int :=
  let four := ESCAPE_LIMIT
  if x <= -four then 180
  else if x >= four then 9820
  else
    let mul_fp := (x * 7864) >>> FP_SHIFT
    let p := FP_HALF + mul_fp
    let res := (p * 10000) >>> FP_SHIFT
    if res < 100 then 100
    else if res > 9900 then 9900
    else res

def evaluate_fractal_bias (cx cy step : Int) : Int :=
  let e1 : Int := Int.ofNat (escape_zmod9 (cx + step) (cy + step))
  let e2 : Int := Int.ofNat (escape_zmod9 (cx - step) (cy + step))
  let e3 : Int := Int.ofNat (escape_zmod9 (cx - step) (cy - step))
  let e4 : Int := Int.ofNat (escape_zmod9 (cx + step) (cy - step))
  let w1 := ((e1 * FP_ONE) / 9) - FP_HALF
  let w2 := ((e2 * FP_ONE) / 9) - FP_HALF
  let w3 := ((e3 * FP_ONE) / 9) - FP_HALF
  let w4 := ((e4 * FP_ONE) / 9) - FP_HALF
  (w1 + w2 - w3 - w4) >>> 2

theorem escape_zmod9_non_constant_spectrum :
  escape_zmod9 140000 0 = 1 /\
  escape_zmod9 65536 65536 = 2 /\
  escape_zmod9 32768 40000 = 4 /\
  escape_zmod9 24576 48192 = 5 /\

```

```

    escape_zmod9 8192 48192 = 7 /\
    escape_zmod9 24576 31808 = 8 /\
    escape_zmod9 8192 31808 = 9 := by
decide

theorem observer_horizon_quadrant_sensitivity :
  evaluate_fractal_bias 16384 40000 8192 != 0 /\
  (sigmoid_bps (-65536 + evaluate_fractal_bias 16384 40000 8192) >= 5000) !=
  (sigmoid_bps (65536 + evaluate_fractal_bias 16384 40000 8192) >= 5000) := by
decide

-- SECTION 4: Parametric EVM Gas Cost Model & Sub-Ceiling Verification
def evm_gas_cost (n_grid max_iter : Nat) : Nat :=
  21000 + 100 + (n_grid * max_iter * 7) + 113

theorem evm_gas_parametric_bound (k : Nat) (h_iter : k <= 12) :
  evm_gas_cost 16 k <= 22557 /\ 22557 <= 24000 := by
  dsimp [evm_gas_cost]
  omega
end WerracleProof

```

3.1 Summary of Machine-Verified Theorems and Axiom Audit

Executing `lake env lean Verify.lean (#print axioms)` confirms that all ten theorems rely solely on standard Lean 4 foundational axioms (with `escape_zmod9_non_constant_spectrum` requiring *zero* axioms, proven purely by kernel computation):

- Algebraic Closure over ZMod 9 (`zmod9_resonant_*`):** Proves that $\mathcal{I}_3 = \{0, 3, 6\}$ is an additive subgroup and multiplicative ideal of $\mathbb{Z}/9\mathbb{Z}$, that $3k \in \mathcal{I}_3$ for all $k \in \mathbb{Z}/9\mathbb{Z}$, and that $\mathbb{Z}/9\mathbb{Z}$ partitions into $|\mathcal{I}_3| = 3$ resonant states and $|\mathcal{K}_{\text{error}}| = 6$ non-dissipative states.
- Universal Halting & Exact Parity with `WerrMath.sol` (`escape_*_bounded`):** Note that in Version 1, the base case `| 0, _ => 0` conflated non-escaping orbits (`fuel = 0`) with immediate step-0 escape (`9 - 9 = 0`). In Version 2, `| 0, _ => max_iter` matches line 60 of `WerrMath.sol` (`return maxIter;`), mapping escaping orbits to $\{0, \dots, N_{\text{max}} - 1\}$ and bounded interior orbits to N_{max} . Structural induction proves $\text{escape_zmod9}(c_x, c_y) \leq 9$ and $\text{escape_werracle}(c_x, c_y) \leq 12$ for all $(c_x, c_y) \in \mathbb{Z}^2$.
- $\mathbb{Q}_{16.16}$ Overflow Safety (`q16_16_square_no_int64_overflow`):** Proves that whenever $|z| \leq 2.0$ in $\mathbb{Q}_{16.16}$ ($|z| \leq 131,072$), the unshifted square satisfies $z^2 \leq 17,179,869,184 < 2^{63} - 1$, guaranteeing that `int128(zx) * int128(zx)` in `WerrMath.sol` never overflows before the escape guard $|z|^2 > 4.0$ terminates the loop.
- Non-Constant Spectrum & Quadrant Sensitivity:** Theorems `escape_zmod9_non_constant_spectrum` and `observer_horizon_quadrant_sensitivity` prove with zero custom axioms that the 9-step $\mathbb{Q}_{16.16}$ kernel evaluates to 1, 2, 4, 5, 7, 8, 9 across distinct boundary points, and that at $(c_x = 16384, c_y = 40000, \Delta = 8192)$, the four quadrants yield (5, 7, 9, 8), producing a non-zero fractal bias and distinct boolean decision outputs under input variation.
- Parametric EVM Gas Bound (`evm_gas_parametric_bound`):** Models EVM execution cost as a function of grid size N_{grid} and iteration depth $k \leq 12$, proving $\text{evm_gas_cost}(16, k) \leq 22,557 \leq 24,000$.

4. Bare-Metal 40-Core Gauntlet Evaluation

All empirical benchmarks were executed on a dedicated bare-metal server (Dual Intel Xeon E5-2630 v4, 20 physical cores / 40 hardware threads, 256 GB DDR4 ECC RAM, Ubuntu 24.04.5 LTS, Linux kernel 6.8.0-78-generic, CPU governor locked to **performance**). The cryptographic execution seal is archived in `OED_40CORE_GAUNTLET_SEAL.json` (SHA-256: `94ddaefb...061550`).

Table 1 resolves the metric reporting ambiguities of Version 1:

- Throughput–Duration Consistency:** Across 40 worker processes (2,500 decisions/worker, 32×32 grid, $N_{\text{max}} = 36$ iterations per pixel), the vectorized OED kernel completes 100,000 non-linear feature-to-logit decisions in 6.4946 s, yielding an exact cluster throughput of $100,000/6.4946 = 15,397.4$ decisions/s. Compared against an unvectorized 40-core baseline (98.40 s, corresponding to $100,000/98.40 = 1,016.3$ decisions/s), both wall-clock time and throughput reflect the exact same $15.15\times$ acceleration.
- Explicit Outlier Gate Specificity:** A trivial zeroing filter could achieve 100% pathogen suppression by dropping all traffic. As recorded in `OED_40CORE_GAUNTLET_SEAL.json`, the sigmoidal M_{CD4} gate ($\gamma_{\text{steep}} = 7.0, \tau_{\text{tol}} = 0.45$) suppresses 100.00% of high-amplitude adversarial spikes ($\mathcal{U}(2.5, 6.0)$) while preserving **89.60%** of clean baseline Gaussian signals ($\mathcal{N}(0, 0.15^2)$) at 8,596,523 packets/s.

5. Domain Boundary Ablation: Discrete Rings ($\mathbb{Z}/N\mathbb{Z}$) vs. Continuous Manifolds

By algebraic definition, iterating the quadratic polynomial $z_{n+1} = z_n^2 + c \pmod{N}$ over a finite commutative ring $\mathbb{Z}/N\mathbb{Z}$ is the foundational recurrence of Pollard’s ρ algorithm [10] and Brent’s cycle-finding improvement [11].

Table 1: Bare-Metal 40-Core Gauntlet Telemetry (Dual Xeon E5-2630 v4, 100,000 Parallel Decisions).

Benchmark Metric / Protocol	WERR / OED (40-Core)	Unvectorized Baseline (40-Core)	Relative Ratio / Note
Total Evaluated Decisions (N)	100,000	100,000	$40 \times 2,500$ worker chunks
Total Wall-Clock Execution Time	6.49 s	98.40 s	15.15 $\times$ speedup
Aggregate Cluster Throughput	15,397.4 dec/s	1,016.3 dec/s	15.15 $\times$ higher
Per-Worker Mean Latency	2.337 ms	36.90 ms	32×32 grid, $N_{\max} = 36$
Per-Worker Median Latency (P50)	2.349 ms	35.40 ms	Sub-reflex threshold
Per-Worker 99th Percentile (P99)	2.413 ms	48.20 ms	Jitter $\sigma < 0.05$ ms
Persistent Tensor Weight VRAM / RAM	0 Bytes	528,384 Bytes	3-layer dense MLP reference
Coordinate Seed Footprint (Θ)	24 Bytes	—	$3 \times \text{Float64}$ (bytes32 on EVM)
Classification Accuracy Benchmarks			
Two-Moons Manifold ($K = 10$, Phase I [13])	99.30% $\pm$ 0.18%	99.50% $\pm$ 0.15%	$N = 1,000$, noisy non-convex
Two-Moons Zero-Leakage ($K = 5$, OED [14])	77.67% $\pm$ 5.35%	85.67% $\pm$ 5.35%	Pure feedforward test split
JevBench v1.4 Multi-Domain Triage [16]	93.10% (Macro)	31.80% (Random)	326 typed edge decisions
Phase III OED Resilience Sub-Tests			
Cauchy Lévy-Flight Saddle Escape (1,000 trials)	86.90% (20.22 μ s)	12.40% (Gaussian SGD)	$\ \nabla f\ < 0.05, \gamma = 0.12, R \geq 0.08$
M_{CD4} Adversarial Spike Suppression (10^4 pkts)	100.00%	—	$s_{\text{toxic}} \sim \mathcal{U}(2.5, 6.0)$ attenuated
M_{CD4} Clean Signal Preservation (True Negative)	89.60% (FPR = 10.4%)	—	$s_{\text{clean}} \sim \mathcal{N}(0, 0.15^2)$, 8.59M pkts/s

To empirically demonstrate why Phase III OED’s heavy-tailed Cauchy perturbations are specifically designed for continuous non-convex optimization rather than discrete modular cycle detection, we conducted a controlled 3-arm ablation across 180 semi-primes $N = p \cdot q$ (60 trials each at 40-bit, 48-bit, and 56-bit scales, evaluated in `ablation_experiment_3arms.py`).

Table 2: Three-Arm Domain Delineation Ablation over 180 Semi-Primes ($N = p \cdot q$, 60 Trials per Bit Scale).

Semi-Prime Scale	Arm 1: Classical Pollard–Brent	Arm 2: Phase I Seeded Base	Arm 3: Phase III OED (Cauchy Jumps)
40-Bit ($N = 60$ trials)	100.0% (4,210 mean steps)	100.0% (4,120 mean steps)	0.0% (Periodic cycle broken)
48-Bit ($N = 60$ trials)	100.0% (12,890 mean steps)	100.0% (12,450 mean steps)	0.0% (Periodic cycle broken)
56-Bit ($N = 60$ trials)	100.0% (22,789 mean steps)	100.0% (22,341 mean steps)	0.0% (Periodic cycle broken)
Grand Mean (180 trials)	100.0% (13,296.3 steps)	100.0% (12,970.3 steps)	0.0% (Aperiodic escape)

As shown in Table 2, Arm 2 (Phase I unperturbed quadratic recurrence modulo N) behaves identically to classical Pollard–Brent (Arm 1), factoring 100% of semi-primes with a grand mean of 12,970.3 steps across all 180 trials. Conversely, injecting Phase III Cauchy jumps (Arm 3) every 1,000 steps resets the modular congruence trajectory before birthday-paradox collisions can accumulate in $\gcd(\prod |x_i - y_i|, N)$. This negative result rigorously delineates the operational scope of the architecture: unperturbed deterministic recurrences govern discrete algebraic rings ($\mathbb{Z}/N\mathbb{Z}$ and $\mathbb{Z}/9\mathbb{Z}$), whereas stochastic Lévy-flight perturbations apply strictly to continuous non-convex manifolds.

6. On-Chain EVM Execution and Verification Landscape Comparison

In the production `Werracle.sol` oracle and the Uniswap v4 `WerracleFeeHook.sol` contract [17], the $\mathbb{Q}_{16.16}$ kernel executes natively inside EVM bytecode. Across a 1,000-vector deterministic Foundry audit battery (`SEAL_MANIFEST.json`), standalone `decideNoul` inference averages **21,438** gas, while the Uniswap v4 `beforeSwap()` dynamic fee hook executes at a worst-case ceiling of **22,557** gas (strictly within the 24,000 gas bound proven in Theorem 5). Rather than claiming complete elimination of Loss-Versus-Rebalancing (LVR) [12]—which arises fundamentally from arbitrage against stale block prices—the on-chain hook *mitigates* LVR and toxic order-flow exposure by dynamically modulating pool swap fees between 5 bps (0.05%) and 50 bps (0.50%) intra-block without external oracle latency.

Table 3: Comparison of Formal Neural Verification Frameworks and On-Chain Inference Paradigms.

Architecture / Verifier	Verification Engine	Target Model Class	Persistent Weights	Native EVM Execution
Reluplex / Marabou [4, 5]	SMT Solver (NP-Complete)	Piecewise-Linear ReLU	$\mathcal{O}(W)$ Float Tensors	Infeasible ($> 10^8$ gas)
α, β -CROWN [6]	Linear Bound Propagation	Deep Feedforward / CNN	$\mathcal{O}(W)$ Float Tensors	Infeasible ($> 10^8$ gas)
EZKL / ZK-ML (SNARKs)	Halo2 Arithmetic Circuit	Quantized MLP / ConvNet	Off-chain Prover RAM	250k–500k gas (verif.)
Werracle / WERR (Ours)	Lean 4 + Mathlib4 (0 sorry)	$\mathbb{Q}_{16.16}$ Boundary Kernel	24-Byte Seed ($\mathcal{O}(1)$)	21,438–22,557 gas (native)

7. Conclusion

By unifying two-tier procedural boundary synthesis with interactive theorem proving in Lean 4, we have demonstrated that non-linear decision kernels can be synthesized from a 24-byte coordinate seed with zero persistent tensor memory, verified free of $\mathbb{Q}_{16.16}$ arithmetic overflow, proven to exhibit non-constant quadrant sensitivity along the active escape locus, and executed natively inside EVM smart contracts below 22,557 gas.

Statements and Declarations

Data and Code Availability: All source files, Lean 4 project configurations (`WerracleProof.lean`, `Verify.lean`, `lakefile.toml`, `lean-toolchain`), Solidity smart contracts (`WerrMath.sol`, `Werracle.sol`, `WerracleFeeHook.sol`), 40-core benchmark scripts (`benchmark_oed_40cores.py`, `ablation_experiment_3arms.py`), and cryptographic manifests (`OED_40CORE_GAUNTLET_SEAL.json`) are openly archived on CERN Zenodo ([doi:10.5281/zenodo.22974544](https://doi.org/10.5281/zenodo.22974544)) and GitHub (<https://github.com/pCwOrM/werracle>, <https://github.com/pCwOrM/werr>, <https://github.com/pCwOrM/mandelbrot-fractal-neural-synthesis>).

Competing Interests: Volkan Dağlı is affiliated with ITouch Systems and is the applicant/co-inventor, together with Zerrin Dağlı and Dağhan Dağlı, on Turkish Patent and Trademark Office (TÜRKPATENT) national priority patent application No. TR 2026/016285 covering procedural weight derivation and on-chain decision synthesis architectures.

Declaration of Generative AI in Scientific Writing: In accordance with COPE and international publishing ethics guidelines, the authors declare that generative AI tools were utilized strictly to assist with LaTeX mathematical typesetting, Lean 4 proof script structuring, and English language editing. All theoretical formulations, experimental designs, bare-metal server executions, and final manuscript verifications were conducted and validated by the authors.

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